IMDB Movie Recommendation System Using Storylines – Project Report

1. Project Title

IMDB Movie Recommendation System Using Storylines

2. Skills Gained

Selenium, Python, Pandas, Streamlit, NLP, TF-IDF, Cosine Similarity, Machine Learning, Data Cleaning, Analysis, Visualization.

3. Domain

Entertainment / Data Analytics / Recommender Systems

4. Problem Statement

The project aims to scrape IMDb 2024 movies, extract storylines, preprocess text using NLP, and recommend similar movies based on storyline similarity using TF-IDF and Cosine Similarity.

5. Business Use Cases

- Storyline-based movie recommendations
- Personalized entertainment suggestions

6. Approach Overview

Data Scraping → NLP Processing → Vectorization (TF-IDF) → Similarity Computation → Streamlit App.

7. Data Scraping (Selenium)

Scraped IMDb 2024 movie titles and storylines using Selenium with scrolling and Load More handling.

8. NLP Processing

Storylines cleaned using stopwords removal, text normalization, and TF-IDF vectorization. Cosine similarity computed.

9. Recommendation Engine

User enters storyline → cleaned → vectorized → similarity scores computed → Top 5 movies returned.

10. Streamlit Interface

Interactive UI allowing storyline input and displaying recommended movie titles and storylines.

11. Dataset Details

CSV containing Movie Title and Storyline. Scraped exclusively from IMDb 2024 listings.

12. Tech Stack

Python, Selenium, Pandas, Streamlit, Scikit-learn, NLTK, TF-IDF, Cosine Similarity.

13. Deliverables

Scraping script, NLP processing script, recommendation script, Streamlit UI, CSV dataset, project report.

14. Conclusion

The system successfully recommends movies based on storyline similarity using NLP and ML techniques.